TERM 1

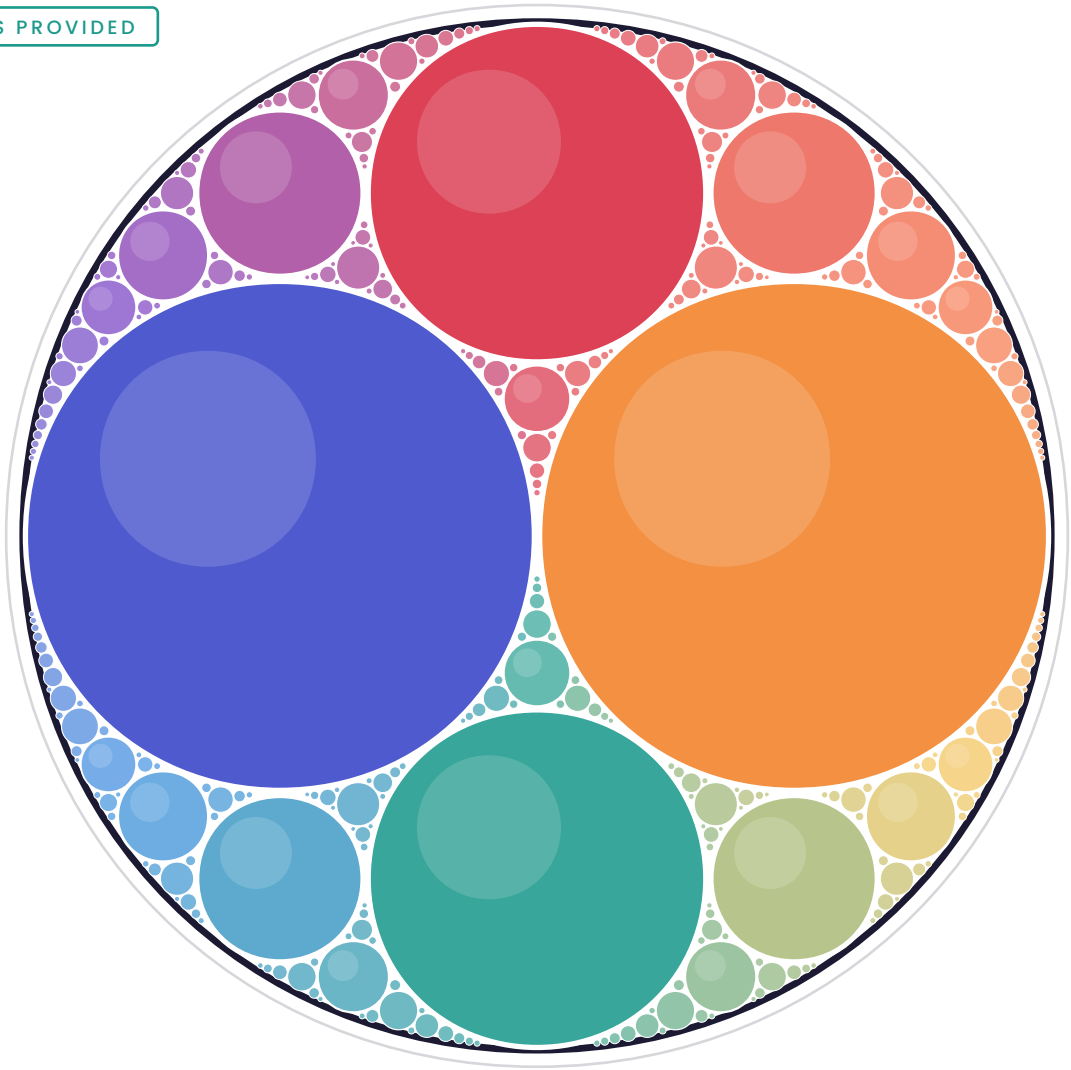
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

APOLLONIAN GASKET
each circle fills a gap between three touching circles; every curvature (1 ÷ radius) is an integer

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

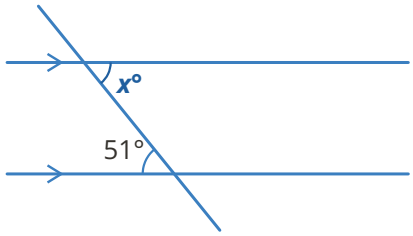
TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION | WORKING | SCORE /5

1 Find the value of x.



FORMULA Alternate angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

2 Evaluate.

$$-14 + 8$$

METHOD Start at the first number on a number line; adding moves right

RULE

WORKING

ANSWER

3 The point $(3, k)$ lies on the line $y = 3x + 5$. Find the value of k.

METHOD A point lies on a line if its coordinates make the rule true

RULE

WORKING

ANSWER

4 Find the mode of these scores.

Scores	6	7	8	5	2	6	6

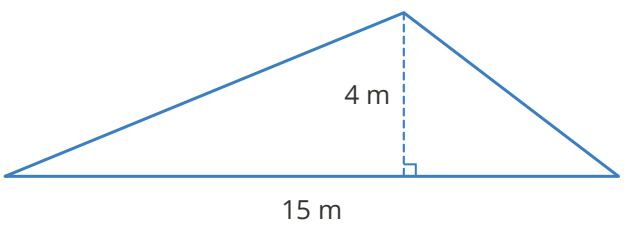
FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

5 Find the area of this triangle.



FORMULA $A = \frac{1}{2}bh$

COPY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the area of this parallelogram.



FORMULA $A = bh$

COPY

SUBSTITUTE

ANSWER

2 Simplify.

$$12x \div 6$$

METHOD Divide the numbers, then cancel pronumerals that are top and bottom

RULE

WORKING

ANSWER

3 $\frac{1}{5}$ of the 45 stalls at the Kingscliff markets sell food. How many stalls is that?

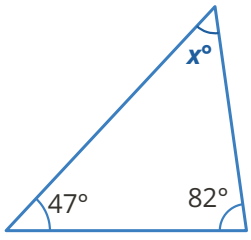
FORMULA Fraction of a quantity = fraction × quantity

COPY

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Angle sum of a triangle = 180°

COPY

SUBSTITUTE

ANSWER

5 A fair die is rolled. Find the probability of rolling a number less than 5.



FORMULA P(E) = number of favourable outcomes ÷ total number of outcomes

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Calculate.

50% of 160 km

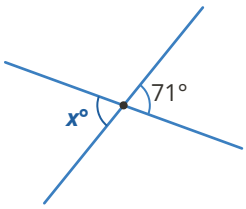
FORMULA Percentage of a quantity = percentage ÷ 100 × quantity

COPY

SUBSTITUTE

ANSWER

2 Find the value of x. Give a reason. REASON



FORMULA Vertically opposite angles are equal

COPY

SUBSTITUTE

REASON

ANSWER

3 Find the value of this expression when $a = 2$ and $b = 9$.

$$ab - 4$$

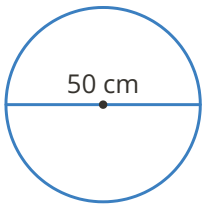
METHOD Replace each pronumeral with its value (use brackets)

RULE

WORKING

ANSWER

4 Find the circumference of this circle, correct to 1 decimal place.



FORMULA $C = \pi d$

COPY

SUBSTITUTE

ANSWER

5 The probability that the counter is red is 0.15. Find the probability that the counter is not red.

FORMULA P(not E) = 1 − P(E)

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 A netball shooter scored 9 goals from 12 shots. What percentage of the shots were goals?

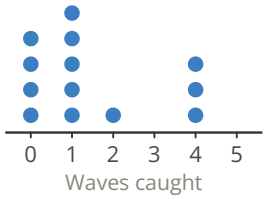
FORMULA Percentage = part ÷ whole × 100%

COPY

SUBSTITUTE

ANSWER

2 The dot plot shows the waves each nipper caught in a session. Find the mode.



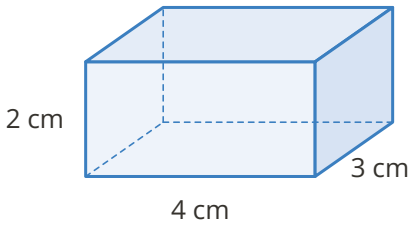
FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

3 Find the volume of this rectangular prism.



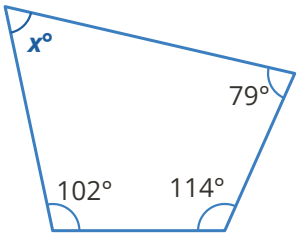
FORMULA $V = lwh$

COPY

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Angle sum of a quadrilateral = 360°

COPY

SUBSTITUTE

ANSWER

5 Is $a = 11$ the solution of this equation? Show how you know. EXPLAIN

$$5a - 9 = 41$$

METHOD Check a solution by substituting it into both sides

RULE

WORKING

EXPLAIN

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 A netball goal shooter scored 16 goals from 20 shots. Find the observed probability of scoring a goal.

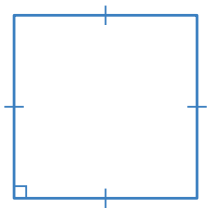
FORMULA Observed probability = frequency ÷ number of trials

COPY

SUBSTITUTE

ANSWER

2 Find the perimeter of this square.



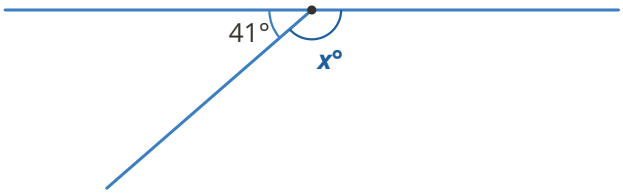
FORMULA $P = 4s$

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



FORMULA Angles on a straight line add to 180°

COPY

SUBSTITUTE

ANSWER

4 Calculate.

$$5.19 \div 1000$$

METHOD × 10, 100, 1000: digits move 1, 2, 3 places left. ÷: they move right

RULE

WORKING

ANSWER

5 Simplify.

$$6 \times 5x$$

METHOD Multiply the numbers, then the pronumerals (in alphabetical order)

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Solve.

$$m + 7 = 21$$

METHOD Do the same to both sides (use the inverse operation)

RULE

WORKING

ANSWER

2 Find the range of these heights.

cm	171	171	174	170	155	156

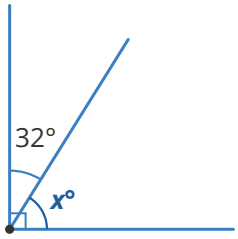
FORMULA Range = highest score − lowest score

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



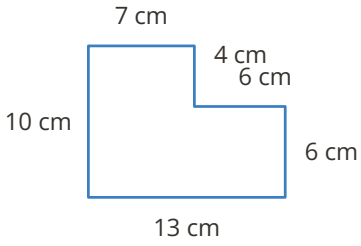
FORMULA Complementary angles add to 90°

COPY

SUBSTITUTE

ANSWER

4 Find the area of this shape.



METHOD Split the shape into rectangles (and triangles), then add their areas

RULE

WORKING

ANSWER

5 Write as a percentage.

$$\frac{2}{5}$$

METHOD Fraction to percentage: fraction × 100%

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

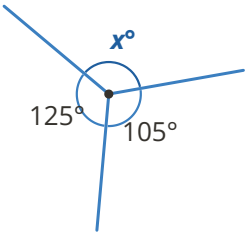
TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the value of x.



FORMULA Angles at a point add to 360°

COPY

SUBSTITUTE

ANSWER

2 Evaluate.

3^3

FORMULA $a^n = a \times a \times \ldots \times a$ (n factors)

COPY

SUBSTITUTE

ANSWER

3 Complete the table for $y = -x + 9$.

x	0	1	2	3
y		8		6

METHOD Substitute each x-value into the rule

RULE

WORKING

ANSWER

4 Find the mean of these waves caught by nippers.

Waves	7	6	6	7	4

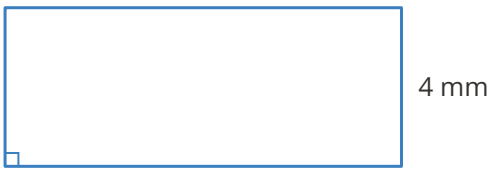
FORMULA Mean = sum of scores ÷ number of scores

COPY

SUBSTITUTE

ANSWER

5 Find the area of this rectangle.



FORMULA $A = lw$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

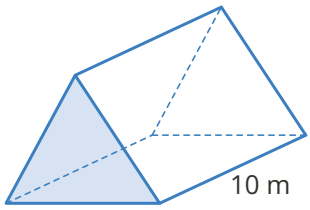
TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The shaded cross-section has an area of 10 m^2. Find the volume of this prism.



FORMULA $V = Ah$

COPY

SUBSTITUTE

ANSWER

2 Sam says $7 + 4 \times 5 = 55$. Explain the mistake and give the correct answer. EXPLAIN

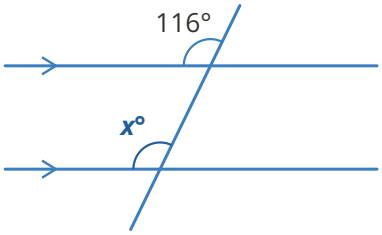
METHOD Brackets, then × and ÷ (left to right), then + and − (left to right)

RULE

WORKING

EXPLAIN

3 Find the value of x.



FORMULA Corresponding angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

4 Find the median of these scores.

Scores	17	4	19	9	13	12	15

FORMULA Median = middle score when the scores are in order (2 middles: halfway)

COPY

SUBSTITUTE

ANSWER

5 Simplify.

$$5a + 4b + 5a - 3b$$

METHOD Only like terms (the same pronumerals) can be added or subtracted

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

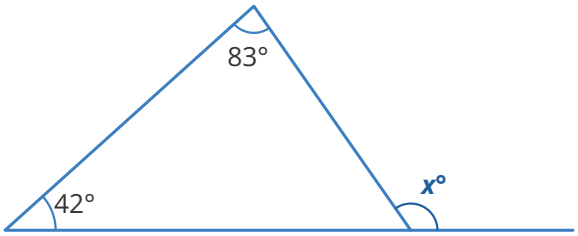
TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the value of x.



FORMULA Exterior angle = sum of the 2 interior opposite angles

COPY

SUBSTITUTE

ANSWER

2 Evaluate.

$$\sqrt{169}$$

METHOD $\sqrt{a}$ is the number that squares to a; $\sqrt[3]{a}$ is the number that cubes to a

RULE

WORKING

ANSWER

3 Convert 5 m^2 to cm^2.

FORMULA 1 cm^2 = 100 mm^2, 1 m^2 = 10 000 cm^2, 1 ha = 10 000 m^2

COPY

SUBSTITUTE

ANSWER

4 The stem-and-leaf plot shows the ages of people at a beach. Find the range.

1	5 9
2	
3	0 2 3

Key: 1 | 5 = 15

FORMULA Range = highest score − lowest score

COPY

SUBSTITUTE

ANSWER

5 Write an algebraic expression for:

double m, then add 3

METHOD Turn words into symbols: more than +, less than −, times ×

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Calculate.

$$2.5 \times 3$$

METHOD Ignore the points, multiply, then count the decimal places back in

RULE

WORKING

ANSWER

2 Convert 1.5 L to cubic centimetres.

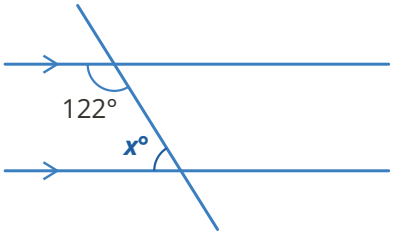
FORMULA 1 cm^3 = 1 mL, 1 L = 1000 mL, 1 m^3 = 1000 L

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



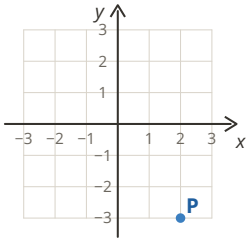
FORMULA Co-interior angles on parallel lines add to 180°

COPY

SUBSTITUTE

ANSWER

4 Write the coordinates of the point P.



METHOD Coordinates (x, y): across first, then up or down

RULE

WORKING

ANSWER

5 A spinner has 4 equal sections. 3 of them are red. Find the probability of landing on red.

FORMULA P(E) = number of favourable outcomes ÷ total number of outcomes

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

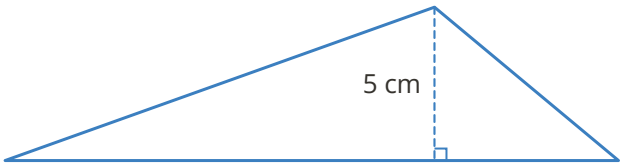
TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the area of this triangle.



FORMULA $A = \frac{1}{2}bh$

COPY

SUBSTITUTE

ANSWER

2 Calculate. Give your answer in simplest form.

$$\frac{4}{8} + \frac{2}{8}$$

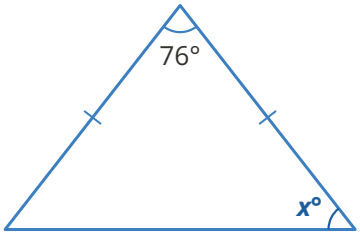
METHOD Same denominators: add or subtract the numerators, then simplify

RULE

WORKING

ANSWER

3 Find the value of x.



FORMULA Isosceles triangle: base angles are equal; angle sum = 180°

COPY

SUBSTITUTE

ANSWER

4 Simplify.

$$12xy \div 3x$$

METHOD Divide the numbers, then cancel pronumerals that are top and bottom

RULE

WORKING

ANSWER

5 Find the mode of these scores.

Scores	7	9	7	3	7	8	1

FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The point $(4, k)$ lies on the line $y = 4x + 4$. Find the value of k.

METHOD A point lies on a line if its coordinates make the rule true

RULE

WORKING

ANSWER

2 Which has the larger area: a 11 m by 4 m rectangle or a square with sides of 7 m? Explain. EXPLAIN

FORMULA $A = lw$ and $A = s^2$

COPY

WORKING

EXPLAIN

3 Round to 1 decimal place.

8.0952

METHOD Look at the next digit: 5 or more, round up; 4 or less, leave it

RULE

WORKING

ANSWER

4 The probability that the bus is late is 80%. Find the probability that the bus is not late.

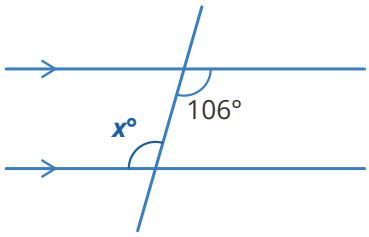
FORMULA P(not E) = 1 − P(E)

COPY

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Alternate angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 For these scores, which is larger: the mean or the median? Explain. EXPLAIN

Scores	8	5	19	2	6

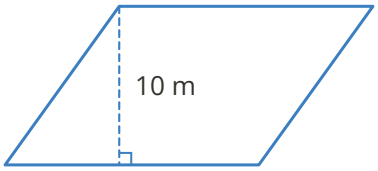
FORMULA Mean = sum of scores ÷ number of scores

COPY

WORKING

EXPLAIN

2 Find the area of this parallelogram.



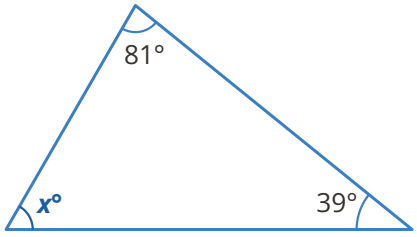
FORMULA $A = bh$

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



FORMULA Angle sum of a triangle = 180°

COPY

SUBSTITUTE

ANSWER

4 Write 98 as a product of its prime factors.

METHOD Use a factor tree: keep splitting until every branch ends in a prime

RULE

WORKING

ANSWER

5 Solve.

$$\frac{p}{3} + 8 = 10$$

METHOD Do the same to both sides (use the inverse operation)

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

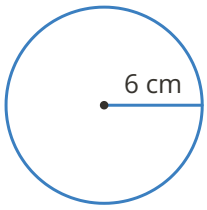
TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Find the circumference of this circle, correct to 1 decimal place.



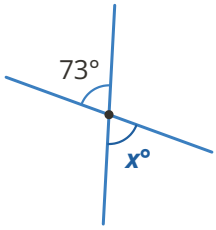
FORMULA $C = 2\pi r$

COPY

SUBSTITUTE

ANSWER

2 Find the value of x.



FORMULA Vertically opposite angles are equal

COPY

SUBSTITUTE

ANSWER

3 Write as a recurring decimal (use dot notation).

$$\frac{5}{6}$$

METHOD Numerator ÷ denominator; put a dot over each repeating digit

RULE

WORKING

ANSWER

4 Find the value of this expression when $m = 9$.

$$3(2m - 1)$$

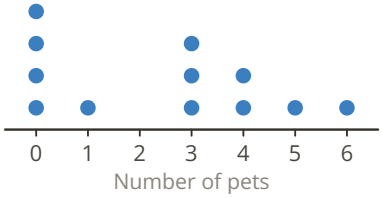
METHOD Replace each pronumeral with its value (use brackets)

RULE

WORKING

ANSWER

5 The dot plot shows the number of pets each student in a class owns. Find the mode.



FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION | WORKING | SCORE /5

1 Decrease 160 L by 15%.

METHOD Find the percentage of the amount, then take it away

RULE

WORKING

ANSWER

2 A die was rolled 60 times. Find the observed probability of rolling a 4.

Number	1	2	3	4	5	6
Frequency	13	7	14	6	13	7

FORMULA Observed probability = frequency ÷ number of trials

COPY

SUBSTITUTE

ANSWER

3 Evaluate.

$$18 \div (-9)$$

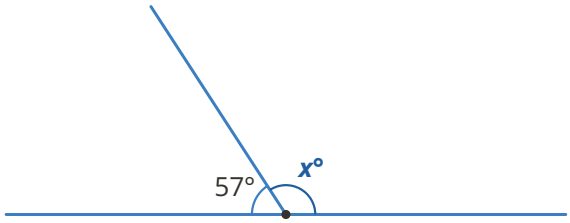
METHOD Same signs: answer positive. Different signs: answer negative

RULE

WORKING

ANSWER

4 Find the value of x.



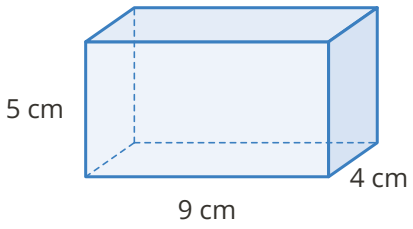
FORMULA Angles on a straight line add to 180°

COPY

SUBSTITUTE

ANSWER

5 Find the volume of this rectangular prism.



FORMULA $V = lwh$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Increase 80 mL by 5%.

METHOD Find the percentage of the amount, then add it on

RULE

WORKING

ANSWER

2 Find the range of these scores.

Scores	10	7	14	9	19	5

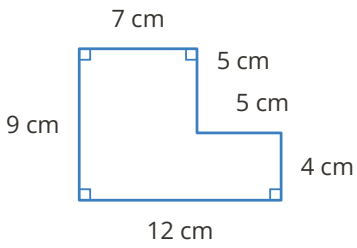
FORMULA Range = highest score − lowest score

COPY

SUBSTITUTE

ANSWER

3 Find the perimeter of this shape.



METHOD Perimeter = the sum of all the side lengths

RULE

WORKING

ANSWER

4 $\frac{2}{3}$ of the 30 students in a class walk to school. How many students is that?

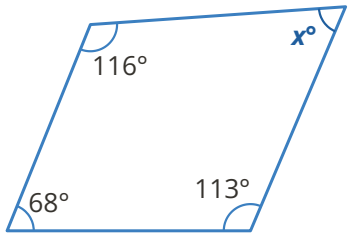
FORMULA Fraction of a quantity = fraction × quantity

COPY

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Angle sum of a quadrilateral = 360°

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The canteen buys a tray of 20 pies for $60 and sells it for $84. Find the profit as a percentage of the cost price.

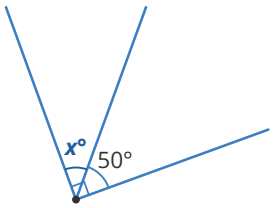
FORMULA Profit (or loss) % = profit (or loss) ÷ cost price × 100%

COPY

SUBSTITUTE

ANSWER

2 Find the value of x. Give a reason. REASON



FORMULA Complementary angles add to 90°

COPY

SUBSTITUTE

REASON

ANSWER

3 Find the mean of these try tallies.

Tries	4	4	1	5	1	3

FORMULA Mean = sum of scores ÷ number of scores

COPY

SUBSTITUTE

ANSWER

4 Calculate.

60% of 45 m

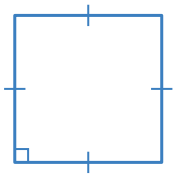
FORMULA Percentage of a quantity = percentage ÷ 100 × quantity

COPY

SUBSTITUTE

ANSWER

5 Find the area of this square.



14 cm

FORMULA $A = s^2$

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 25% of a number is 20. What is the number?

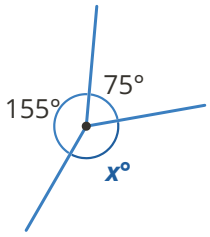
METHOD Unitary method: find 1% (or 10%) first, then multiply to get 100%

RULE

WORKING

ANSWER

2 Find the value of *x*.



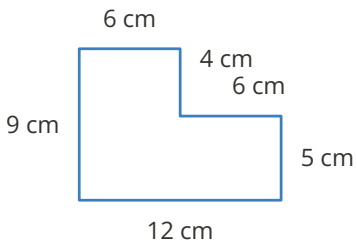
FORMULA Angles at a point add to 360°

COPY

SUBSTITUTE

ANSWER

3 Find the area of this shape.



METHOD Split the shape into rectangles (and triangles), then add their areas

RULE

WORKING

ANSWER

4 Calculate.

$$0.25 \div 10$$

METHOD × 10, 100, 1000: digits move 1, 2, 3 places left. ÷: they move right

RULE

WORKING

ANSWER

5 Find the median of these goal tallies.

Goals	15	9	19	4	3	7

FORMULA Median = middle score when the scores are in order (2 middles: halfway)

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$9^9 \div 9^4$$

FORMULA $a^m \div a^n = a^{m-n}$

COPY

SUBSTITUTE

ANSWER

2 The Kingscliff touch footy team won 14 of its 25 games. What percentage of its games did it win?

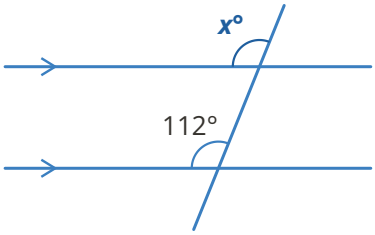
FORMULA Percentage = part ÷ whole × 100%

COPY

SUBSTITUTE

ANSWER

3 Find the value of x.



FORMULA Corresponding angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

4 A letter is chosen at random from this word. Find P(U).

S E A G U L L

FORMULA P(E) = number of favourable outcomes ÷ total number of outcomes

COPY

SUBSTITUTE

ANSWER

5 Solve.

$$8y = 48$$

METHOD Do the same to both sides (use the inverse operation)

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 The price of a skateboard is $170 before GST. Add 10% GST to find the price including GST.

FORMULA GST = 10% of the price; price incl. GST = price + GST

COPY

SUBSTITUTE

ANSWER

2 Simplify. Write your answer as a single power.

$$(5^3)^2$$

FORMULA $(a^m)^n = a^{mn}$

COPY

SUBSTITUTE

ANSWER

3 The stem-and-leaf plot shows the ages of people at a beach. Find the median.

3	0 0 4 5
4	0 2
5	5 6
6	5 9 9

Key: 3 | 0 = 30

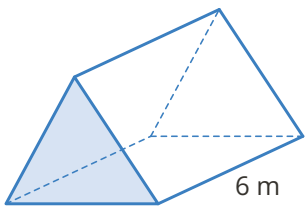
FORMULA Median = middle score when the scores are in order

COPY

SUBSTITUTE

ANSWER

4 The shaded cross-section has an area of 12 m^2. Find the volume of this prism.



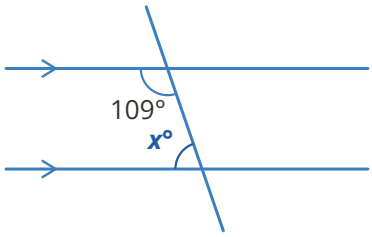
FORMULA $V = Ah$

COPY

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Co-interior angles on parallel lines add to 180°

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Zoe says $5^3 \times 5^5 = 25^8$. Explain the mistake. EXPLAIN

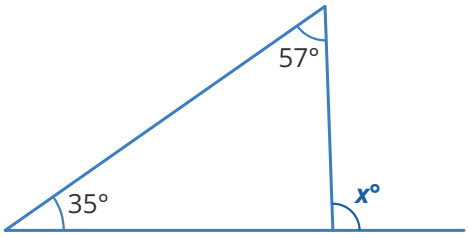
FORMULA $a^m \times a^n = a^{m+n}$

RULE

WORKING

EXPLAIN

2 Find the value of x.



FORMULA Exterior angle = sum of the 2 interior opposite angles

COPY

SUBSTITUTE

ANSWER

3 Find the mode of these scores.

Scores	4	8	3	4	4	5	9	8	1

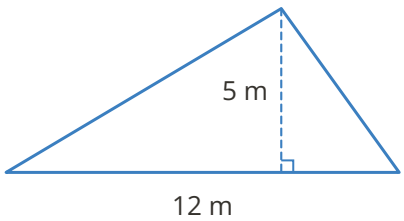
FORMULA Mode = the most common score

COPY

SUBSTITUTE

ANSWER

4 Find the area of this triangle.



FORMULA $A = \frac{1}{2}bh$

COPY

SUBSTITUTE

ANSWER

5 Simplify.

$2 \times 5m$

METHOD Multiply the numbers, then the pronumerals (in alphabetical order)

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Write 72 as a product of prime factors in index form.

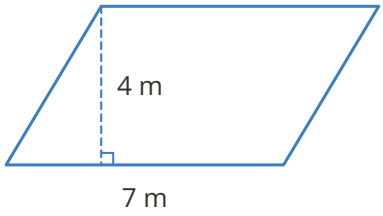
METHOD Factor tree to primes, then write repeated primes as powers

RULE

WORKING

ANSWER

2 Find the area of this parallelogram.



FORMULA $A = bh$

COPY

SUBSTITUTE

ANSWER

3 Write as a fraction in simplest form.

0.6

METHOD Decimal to fraction: write over 10, 100 or 1000, then simplify

RULE

WORKING

ANSWER

4 Does the point (4, 19) lie on the line $y = 3x + 6$? Explain. EXPLAIN

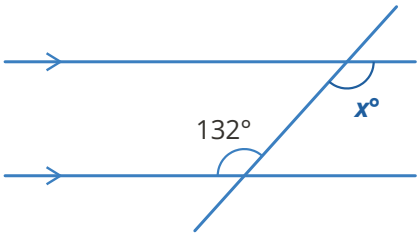
METHOD A point lies on a line if its coordinates make the rule true

RULE

WORKING

EXPLAIN

5 Find the value of x.



FORMULA Alternate angles on parallel lines are equal

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Evaluate.

$$9^2 + 11^0$$

FORMULA $a^0 = 1$ (any non-zero number to the power 0 is 1)

COPY

SUBSTITUTE

ANSWER

2 The number of nippers changed from 60 to 36. Find the percentage decrease.

FORMULA Percentage change = change ÷ original amount × 100%

COPY

SUBSTITUTE

ANSWER

3 Evaluate.

$$10^4$$

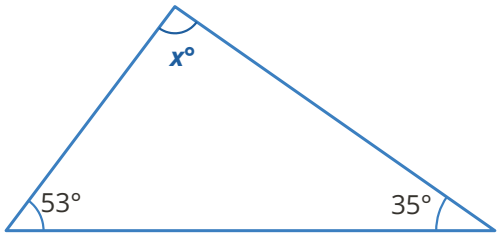
FORMULA $a^n = a \times a \times \ldots \times a$ (n factors)

COPY

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Angle sum of a triangle = 180°

COPY

SUBSTITUTE

ANSWER

5 Complete the table for $y = -x + 6$.

x	0	1	2	3
y		5		3

METHOD Substitute each x-value into the rule

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$5^6 \div 5^2$$

FORMULA $a^m \div a^n = a^{m-n}$

COPY

SUBSTITUTE

ANSWER

2 Simplify.

$$14p - 7p$$

METHOD Only like terms (the same pronumerals) can be added or subtracted

RULE

WORKING

ANSWER

3 The probability that it rains tomorrow is $\frac{1}{4}$.

Find the probability that it does not rain tomorrow.

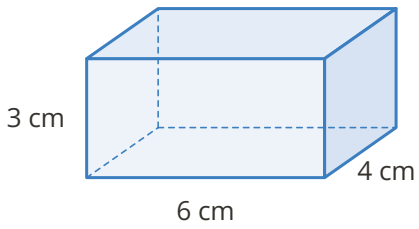
FORMULA P(not E) = 1 − P(E)

COPY

SUBSTITUTE

ANSWER

4 Find the volume of this rectangular prism.



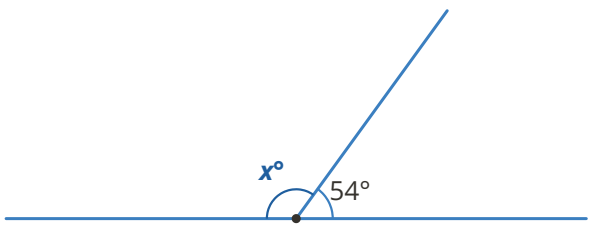
FORMULA $V = lwh$

COPY

SUBSTITUTE

ANSWER

5 Find the value of x.



FORMULA Angles on a straight line add to 180°

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$(9^3)^3$$

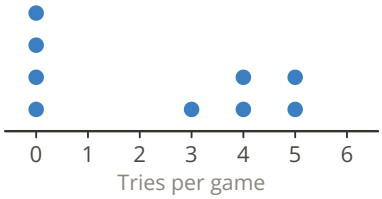
FORMULA $(a^m)^n = a^{mn}$

COPY

SUBSTITUTE

ANSWER

2 The dot plot shows the tries a rugby league team scored in each game. Find the median.



FORMULA Median = middle score when the scores are in order

COPY

SUBSTITUTE

ANSWER

3 Evaluate.

$$\sqrt{144}$$

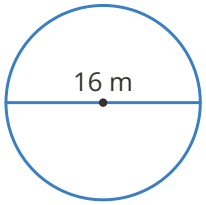
METHOD $\sqrt{a}$ is the number that squares to a; $\sqrt[3]{a}$ is the number that cubes to a

RULE

WORKING

ANSWER

4 Find the circumference of this circle, correct to 1 decimal place.



FORMULA $C = \pi d$

COPY

SUBSTITUTE

ANSWER

5 Simplify.

$$\frac{8x}{2}$$

METHOD Divide the numbers, then cancel pronumerals that are top and bottom

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 A surfboard costs $460. It is on sale at 25% off. Find the sale price.

METHOD Discount = % × price; sale price = price − discount

RULE

WORKING

ANSWER

2 Simplify. Write your answer as a single power.

$$5^2 \times 5^2$$

FORMULA $a^m \times a^n = a^{m+n}$

COPY

SUBSTITUTE

ANSWER

3 A die was rolled 60 times. Find the observed probability of rolling a 6.

Number	1	2	3	4	5	6
Frequency	8	11	9	13	6	13

FORMULA Observed probability = frequency ÷ number of trials

COPY

SUBSTITUTE

ANSWER

4 Sam says 7 + 12 × 9 = 171. Explain the mistake and give the correct answer. EXPLAIN

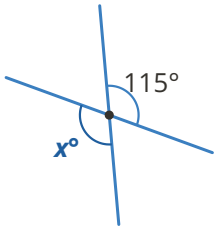
METHOD Brackets, then × and ÷ (left to right), then + and − (left to right)

RULE

WORKING

EXPLAIN

5 Find the value of x.



FORMULA Vertically opposite angles are equal

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Write 54 as a product of prime factors in index form.

METHOD Factor tree to primes, then write repeated primes as powers

RULE

WORKING

ANSWER

2 Find the range of these maximum temperatures.

°C	22	17	21	18	24	18

FORMULA Range = highest score − lowest score

COPY

SUBSTITUTE

ANSWER

3 Convert 1250 mL to litres.

FORMULA 1 cm^3 = 1 mL, 1 L = 1000 mL, 1 m^3 = 1000 L

COPY

SUBSTITUTE

ANSWER

4 Find the value of this expression when $m = 5$ and $n = 5$.

$$5m - 2n$$

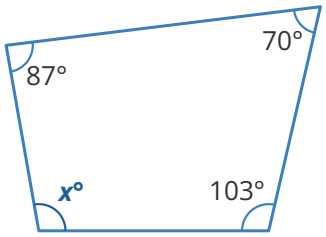
METHOD Replace each pronumeral with its value (use brackets)

RULE

WORKING

ANSWER

5 Find the value of x.



FORMULA Angle sum of a quadrilateral = 360°

COPY

SUBSTITUTE

ANSWER

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ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$9^7 \div 9^2$$

FORMULA $a^m \div a^n = a^{m-n}$

COPY

SUBSTITUTE

ANSWER

2 Find the mean of these goal tallies.

Goals	8	3	4	3	12

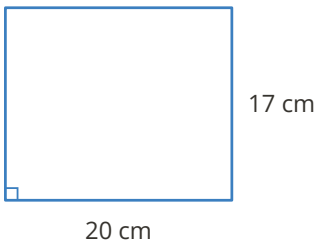
FORMULA Mean = sum of scores ÷ number of scores

COPY

SUBSTITUTE

ANSWER

3 Find the perimeter of this rectangle.



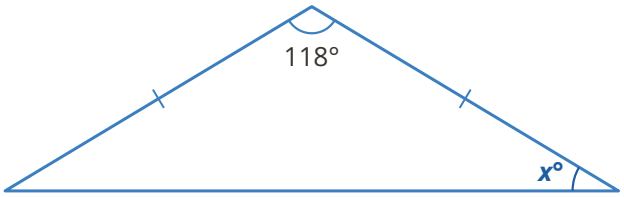
FORMULA $P = 2(l + w)$

COPY

SUBSTITUTE

ANSWER

4 Find the value of x.



FORMULA Isosceles triangle: base angles are equal; angle sum = 180°

COPY

SUBSTITUTE

ANSWER

5 Evaluate.

$$6 - 16$$

METHOD Subtracting moves left on a number line

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 Simplify. Write your answer as a single power.

$$(6^4)^4$$

FORMULA $(a^m)^n = a^{mn}$

COPY

SUBSTITUTE

ANSWER

2 Decrease 60 km by 50%.

METHOD Find the percentage of the amount, then take it away

RULE

WORKING

ANSWER

3 One marble is chosen at random from this bag. Find the probability that it is black.

Colour	White	Red	Black
Marbles	1	8	4

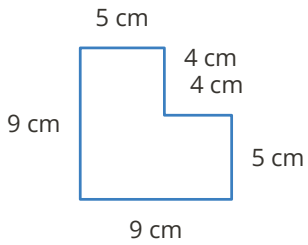
FORMULA P(E) = number of favourable outcomes ÷ total number of outcomes

COPY

SUBSTITUTE

ANSWER

4 Find the area of this shape.



METHOD Split the shape into rectangles (and triangles), then add their areas

RULE

WORKING

ANSWER

5 Calculate.

$$0.5 \times 0.7$$

METHOD Ignore the points, multiply, then count the decimal places back in

RULE

WORKING

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 1 Copy the formula or rule, then use it to answer.

NAME _____

DATE _____

QUESTION	WORKING	SCORE /5

1 A triangle has sides 12 m, 16 m and 20 m. Is it right-angled? Explain. EXPLAIN

FORMULA Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

RULE

WORKING

EXPLAIN

2 Calculate. Give your answer in simplest form.

$$\frac{3}{9} + \frac{5}{9}$$

METHOD Same denominators: add or subtract the numerators, then simplify

RULE

WORKING

ANSWER

3 Find the median of these test marks.

Marks	65	61	54	66	62

FORMULA Median = middle score when the scores are in order (2 middles: halfway)

COPY

SUBSTITUTE

ANSWER

4 Write an algebraic expression for:

5 less than x

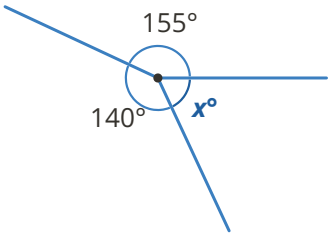
METHOD Turn words into symbols: more than +, less than −, times ×

RULE

WORKING

ANSWER

5 Find the value of x.



FORMULA Angles at a point add to 360°

COPY

SUBSTITUTE

ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Formula · Reference 1

TERM 1 Every formula and rule used in this booklet.

NUMBER

FRACTIONS AND PERCENTAGES OF A QUANTITY

Fraction of a quantity = fraction × quantity

Percentage of a quantity = percentage ÷ 100 × quantity

Percentage = part ÷ whole × 100%

INTEGERS

Start at the first number on a number line; adding moves right

Same signs: answer positive. Different signs: answer negative

Subtracting moves left on a number line

CONVERTING FRACTIONS, DECIMALS AND PERCENTAGES

Fraction to percentage: fraction × 100%

Decimal to fraction: write over 10, 100 or 1000, then simplify

POWERS AND ROOTS

$a^n = a \times a \times \ldots \times a$ (n factors)

$\sqrt{a}$ is the number that squares to a; $\sqrt[3]{a}$ is the number that cubes to a

ADDING, SUBTRACTING AND MULTIPLYING FRACTIONS

Same denominators: add or subtract the numerators, then simplify

MULTIPLYING AND DIVIDING BY 10, 100, 1000

× 10, 100, 1000: digits move 1, 2, 3 places left. ÷: they move right

OPERATIONS WITH DECIMALS

Ignore the points, multiply, then count the decimal places back in

ORDER OF OPERATIONS

Brackets, then × and ÷ (left to right), then + and − (left to right)

PRIME FACTORS (FACTOR TREE)

Use a factor tree: keep splitting until every branch ends in a prime

RECURRING DECIMALS

Numerator ÷ denominator; put a dot over each repeating digit

ROUNDING DECIMALS

Look at the next digit: 5 or more, round up; 4 or less, leave it

INDEX LAWS · YEAR 8

$a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$

$a^m \times a^n = a^{m+n}$

$a^0 = 1$ (any non-zero number to the power 0 is 1)

PERCENTAGE INCREASE AND DECREASE · YEAR 8

Find the percentage of the amount, then take it away

Find the percentage of the amount, then add it on

Percentage change = change ÷ original amount × 100%

Discount = % × price; sale price = price − discount

PROFIT, LOSS AND GST · YEAR 8

Profit (or loss) % = profit (or loss) ÷ cost price × 100%

GST = 10% of the price; price incl. GST = price + GST

PRIME FACTORS IN INDEX FORM · YEAR 8

Factor tree to primes, then write repeated primes as powers

THE UNITARY METHOD · YEAR 8

Unitary method: find 1% (or 10%) first, then multiply to get 100%

ALGEBRA AND EQUATIONS

LINEAR RELATIONSHIPS

A point lies on a line if its coordinates make the rule true

Substitute each x-value into the rule

Coordinates (x, y): across first, then up or down

MULTIPLYING AND DIVIDING TERMS

Divide the numbers, then cancel pronumerals that are top and bottom

Multiply the numbers, then the pronumerals (in alphabetical order)

SOLVING EQUATIONS

Check a solution by substituting it into both sides

Do the same to both sides (use the inverse operation)

LIKE TERMS

Only like terms (the same pronumerals) can be added or subtracted

SUBSTITUTION

Replace each pronumeral with its value (use brackets)

WRITING ALGEBRAIC EXPRESSIONS

Turn words into symbols: more than +, less than −, times ×

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Formula · Reference 2

TERM 1 Every formula and rule used in this booklet (continued).

MEASUREMENT

AREA OF RECTANGLES AND SQUARES

$A = lw$

$A = lw$ and $A = s^2$

$A = s^2$

PERIMETER

$P = 4s$

Perimeter = the sum of all the side lengths

$P = 2(l + w)$

CIRCUMFERENCE OF A CIRCLE

$C = \pi d$

$C = 2\pi r$

UNITS OF AREA, VOLUME AND CAPACITY

1 cm^2 = 100 mm^2, 1 m^2 = 10 000 cm^2, 1 ha = 10 000 m^2

1 cm^3 = 1 mL, 1 L = 1000 mL, 1 m^3 = 1000 L

VOLUME OF PRISMS

$V = lwh$

$V = Ah$

AREA OF A COMPOSITE SHAPE

Split the shape into rectangles (and triangles), then add their areas

AREA OF A PARALLELOGRAM

$A = bh$

AREA OF A TRIANGLE

$A = \frac{1}{2}bh$

PYTHAGORAS' THEOREM · YEAR 8

Right-angled if (longest side)^2 = sum of the squares of the other 2 sides

GEOMETRY: ANGLES

ANGLES AT A POINT AND ON A LINE

Vertically opposite angles are equal

Angles on a straight line add to 180°

Complementary angles add to 90°

Angles at a point add to 360°

TRIANGLES AND QUADRILATERALS

Angle sum of a triangle = 180°

Angle sum of a quadrilateral = 360°

Exterior angle = sum of the 2 interior opposite angles

Isosceles triangle: base angles are equal; angle sum = 180°

ANGLES ON PARALLEL LINES

Alternate angles on parallel lines are equal

Corresponding angles on parallel lines are equal

Co-interior angles on parallel lines add to 180°

STATISTICS AND PROBABILITY

MEAN, MEDIAN, MODE AND RANGE

Mode = the most common score

Range = highest score − lowest score

Mean = sum of scores ÷ number of scores

Median = middle score when the scores are in order (2 middles: halfway)

Median = middle score when the scores are in order

PROBABILITY

P(E) = number of favourable outcomes ÷ total number of outcomes

P(not E) = 1 − P(E)

Observed probability = frequency ÷ number of trials

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

My · Tracker

TERM 1 Write your score after each session.

NAME _____

DATE _____

WEEK	SESSION 1	SESSION 2	SESSION 3	WEEK TOTAL
Week 1	/ 5	/ 5	/ 5	/ 15
Week 2	/ 5	/ 5	/ 5	/ 15
Week 3	/ 5	/ 5	/ 5	/ 15
Week 4	/ 5	/ 5	/ 5	/ 15
Week 5	/ 5	/ 5	/ 5	/ 15
Week 6	/ 5	/ 5	/ 5	/ 15
Week 7	/ 5	/ 5	/ 5	/ 15
Week 8	/ 5	/ 5	/ 5	/ 15
Week 9	/ 5	/ 5	/ 5	/ 15
Week 10	/ 5	/ 5	/ 5	/ 15
TERM TOTAL				**/ 150**

OUTCOMES IN THIS BOOKLET · NSW MATHEMATICS K–10 (2022), STAGE 4

MA4-INT-C-01	compares, orders and calculates with integers to solve problems	NCM 7
MA4-FRC-C-01	represents and operates with fractions, decimals and percentages to solve problems	NCM 7, NCM 8
MA4-IND-C-01	operates with primes and roots, positive-integer and zero indices involving numerical bases and establishes the relevant index laws	NCM 7, NCM 8
MA4-ALG-C-01	generalises number properties to operate with algebraic expressions including expansion and factorisation	NCM 7
MA4-EQU-C-01	solves linear equations of up to 2 steps and quadratic equations of the form $ax^2 = c$	NCM 7
MA4-LIN-C-01	creates and displays number patterns and finds graphical solutions to problems involving linear relationships	NCM 7
MA4-LEN-C-01	applies knowledge of the perimeter of plane shapes and the circumference of circles to solve problems	NCM 7
MA4-ARE-C-01	applies knowledge of area and composite area involving triangles, quadrilaterals and circles to solve problems	NCM 7
MA4-VOL-C-01	applies knowledge of volume and capacity to solve problems involving right prisms and cylinders	NCM 7
MA4-PYT-C-01	applies Pythagoras' theorem to solve problems in various contexts	NCM 8
MA4-ANG-C-01	applies angle relationships to solve problems, including those related to transversals on sets of parallel lines	NCM 7
MA4-GEO-C-01	identifies and applies the properties of triangles and quadrilaterals to solve problems	NCM 7
MA4-DAT-C-02	analyses simple datasets using measures of centre, range and shape of the data	NCM 7
MA4-PRO-C-01	solves problems involving the probabilities of simple chance experiments	NCM 7
MAO-WM-01	Working mathematically: the weekly reasoning questions (marked EXPLAIN or REASON) ask students to explain, check or justify.	

Year 8 units in these warm-ups (each starts once its teaching ends in the 2027 Year 8 mixed ability program): Rational numbers, percentages & financial maths from T1 Wk 5; Index laws from T1 Wk 7; Pythagoras' theorem from T1 Wk 10. All other questions revise Year 7 (NCM 7 Ch 1–11). Ratios, rates and time (NCM 7 Ch 12) are held back until Year 8 teaches them.